CivicFix

From Citizen Complaints to Verified Civic Resolution

ChronoForge 2026 · NITK Surathkal
Theme: **Cities of the Future**
Track: **Agentic AI**

TEAM NAME	TEAM LEADER	MEMBER	INSTITUTION
BitBros	Vikash Saini (241CV256)	Rishi Kumar (241CV144)	NITK Surathkal

GitHub repository: github.com/vikash-saini-000/civicfix

1. Problem Statement and Significance

Urban residents constantly encounter potholes, overflowing garbage bins, broken streetlights, water leaks and damaged public infrastructure. Reporting is only the first step: complaints are **duplicated**, **misclassified**, routed to the **wrong department**, or left with **no visible progress**. Municipal teams struggle to prioritise with limited staff, and complaints are often closed without any proof that the work was actually done.

Existing portals (e.g., FixMyStreet, SeeClickFix, Swachhata-MoHUA, BBMP Sahaaya) mainly *record and route* reports. None of them coordinates the full loop: interpret → deduplicate → prioritise → assign → track → **verify**. CivicFix closes that gap with an agentic AI that turns unstructured complaints into prioritised, trackable and **verified** work orders, reducing administrative overhead and making city services accountable.

Target users: citizens; municipal administrators; public-works, sanitation, electrical and water departments; civic-service coordinators. A natural pilot setting is a mid-sized city corporation such as Mangaluru.

2. Proposed Solution and Key Features

CivicFix has three parts: a **citizen reporting app**, an **AI coordination agent** that acts through controlled backend tools, and an **administrator dashboard** with human-review queues.

A. Intelligent Complaint Intake

Citizens submit a free-text description (English, Hindi or Kannada), a map pin and an optional photo. The agent extracts the category, severity cues and landmarks, asks follow-up questions for missing details, and links likely duplicates. Grouped reports become stronger evidence of an issue's scale.

B. Explainable Prioritisation

Priority (P1–P4) is based on hazard type, severity signals, report count, nearby sensitive locations (schools, hospitals, main roads) and how long the issue has been open. Every decision comes with a written rationale. Low-confidence or sensitive cases go to a human instead of being guessed.

C. Agentic Department Assignment

The agent *plans* and *acts*: it queries simulated department responsibilities, ward coverage, staff availability and workload, then creates a structured work order with an SLA deadline, logging every tool call and the reason for it.

D. End-to-End Tracking

Citizens see each stage in real time. Administrators see a live map, department workloads, overdue SLAs and recurring hotspots (e.g., the same road reported for potholes every monsoon).

Reported → Under Review → Assigned → In Progress → Resolution Submitted → Verified & Resolved | any stage → Flagged for Human Review

E. Closed-Loop Resolution Verification and Audit Trail — our key differentiator

When a department marks work as done, it uploads an "after" photo. The verifier agent (1) uses a vision model to compare the before and after photos for the same issue, (2) checks that the photo's GPS location is within ~50 m of the complaint and its timestamp is after assignment, and (3) checks that the claimed fix matches the issue category. Only consistent evidence closes the complaint. Anything unclear or suspicious is **flagged, never auto-accepted**. A complete audit trail records every agent decision, tool call, status change and human approval, and the citizen can confirm or reopen the complaint.

3. Technical Approach and Architecture

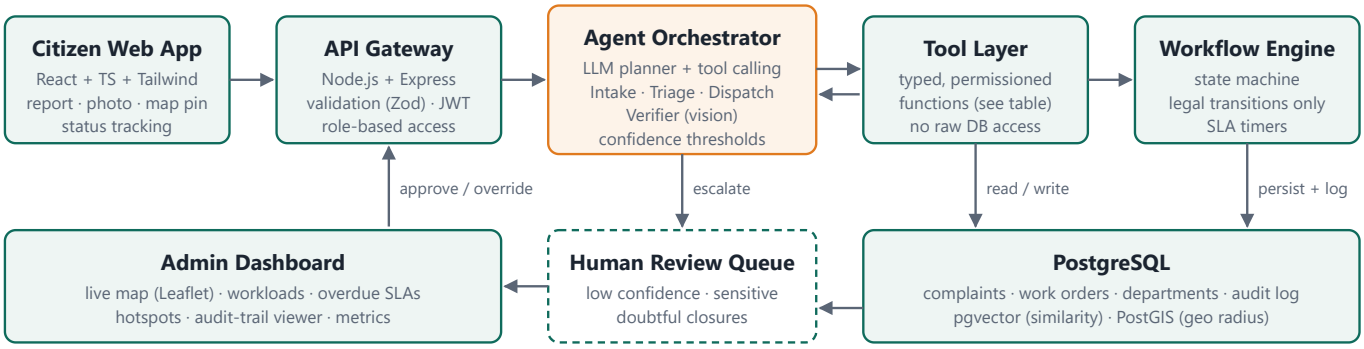


Figure 1 — CivicFix architecture. The LLM proposes actions; only validated tools and the workflow engine change state.

Agent tools and guardrails

Tool	What it does	Guardrail
`classify_complaint`	Category, severity cues, landmarks, missing fields	Confidence < 0.7 → human review
`find_similar_complaints`	Text-embedding similarity + PostGIS radius (~100 m) + 14-day window	Merge only above a similarity threshold; else "possible duplicate" link
`get_department_capacity`	Responsibility map, ward coverage, staff on duty, open workload	Read-only
`set_priority`	Assigns P1–P4 with a stored written rationale	P1 and sensitive categories require admin confirmation
`create_work_order`	Assigns department and team, sets SLA deadline	Workflow engine rejects invalid status transitions
`verify_resolution`	Vision before/after comparison, GPS + timestamp consistency	Can only *approve* or *flag*; never closes when unsure
`request_human_review`	Escalates with reason and evidence bundle	Always allowed

Technology stack

- **Frontend:** React, TypeScript, Tailwind CSS, Leaflet + OpenStreetMap for maps and hotspot heatmaps.
- **Backend:** Node.js, Express, REST APIs, Zod schema validation, JWT authentication, role-based access (citizen / department / admin).
- **Data:** PostgreSQL with **pgvector** (duplicate similarity) and **PostGIS** (geo queries); an append-only audit-log table.
- **AI layer:** a tool-calling LLM (e.g., Claude Sonnet) for planning and classification, a multimodal model for photo verification, and an embedding model for similarity. A model-agnostic adapter lets us swap providers or run an open model.
- **Quality:** unit tests for tools and state transitions, a scenario test harness, and seeded simulated city data (wards, departments, staff).

Agent loop (per complaint)

1 Citizen submits → **2** API validates and stores (status *Reported*) → **3** agent classifies and searches for duplicates → **4** agent fetches department, workload and location context → **5** agent proposes a priority and assignment, then executes the permitted tool calls → **6** workflow engine validates each transition and writes the audit log → **7** on completion, the verifier checks the evidence and either closes the complaint or escalates it to an administrator. If a tool fails, the agent retries with backoff and then falls back to the human queue, so no complaint is silently dropped.

4. Innovation and Differentiation

Capability	Typical complaint portals	CivicFix
Complaint understanding	Fixed category dropdowns	**Free text, multilingual, follow-up questions**
Duplicate handling	Manual or none	**Semantic + geo + time clustering**
Prioritisation	First-come or manual	**Evidence-based, with written rationale**
Assignment	Static routing rules	**Agent plans using capacity and workload**
Proof of resolution	Status set by department	**Vision + GPS/time verification, flags doubtful closures**
Accountability	Basic status history	**Full audit trail of AI decisions and human approvals**

CivicFix is **action-oriented AI with verifiable outcomes**: the agent reasons, plans and executes through real tools under **constrained autonomy**. Humans stay in control of high-impact decisions, and every outcome is checked rather than assumed. This maps directly onto the track's criteria: autonomy, tool integration, planning, reliability and outcome verification.

5. Feasibility and 24-Hour Implementation Plan

Both members are comfortable with the JavaScript/TypeScript stack, so we can focus on the agent workflow rather than learning tools. Simulated civic data (wards, departments, staff rosters, ~150 seeded complaints) makes the demo reproducible and independent of real municipal systems. We will scope tightly:

Hours	Milestone
0–5	Schema + seed data, Express API, complaint submission UI with map pin and photo upload
5–12	Agent orchestrator, tool layer, classification, duplicate search (pgvector + PostGIS), prioritisation, work-order creation
12–17	Workflow state machine, audit log, human-review queue, verification agent (photo + GPS/time checks)
17–21	Admin dashboard (map, workloads, SLAs), citizen tracking view, error handling and fallbacks
21–24	Run evaluation scenarios vs. baseline, fix failures, record backup demo video, documentation

Must-have demo: complaint → agent triage → work order → tracking → verified closure → audit trail. **Stretch goals:** Kannada/Hindi voice input, WhatsApp intake, hotspot prediction, citizen notifications.

6. Expected Impact, Evaluation and Scalability

Impact

Faster triage, fewer duplicate work orders, fewer false closures, and transparent progress that rebuilds citizen trust. The same engine fits campus facilities, residential communities and public-works desks.

Evaluation (vs. rule-based baseline)

- Classification and assignment accuracy
- Duplicate-detection precision and recall
- Time from submission to work order
- Share of workflow actions completed
- Verification accuracy on planted fake or insufficient "after" photos

Scalability

Departments, categories and SLAs are configuration, not code, so new cities can be onboarded without rewriting. The stateless API scales horizontally, the multilingual intake extends to new languages, and the tool layer can later connect to real municipal ERPs.

Test set: ~150 labelled synthetic complaints across 8 categories, ~20% planted duplicates and ~15 fake or insufficient resolution photos, all run through both CivicFix and a keyword-rule baseline. We will report the results we actually measure, not assumed improvements.

7. Responsible AI

- **Privacy:** automatic blurring of faces and vehicle number plates in uploads; minimal personal data; reporter identity hidden from departments.
- **Human in the loop:** confidence thresholds, mandatory review for sensitive categories (e.g., electrical hazards, public safety), and admin override of any agent decision.
- **Safety and abuse:** least-privilege tools, rate limiting and spam detection, and an append-only audit log for accountability.

Conclusion. In the cities of the future, reporting a civic problem should start a coordinated, accountable process, not just add another row to a database. CivicFix combines agentic planning, controlled tool execution, transparent decisions and verified resolution to show how AI can make cities measurably more responsive to the people who live in them.